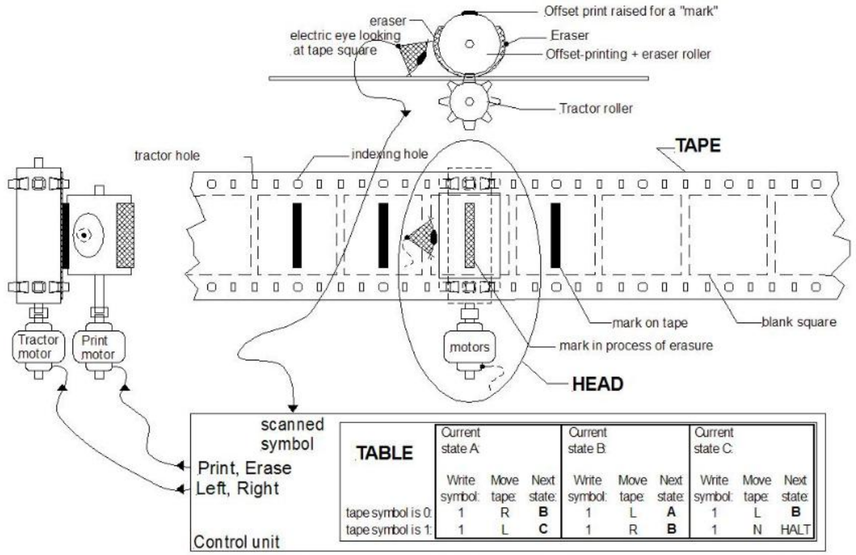
1. Turing Machine

The Universal Logic

A theoretical mathematical model of a device that manipulates symbols on a strip of tape according to a table of rules. It serves as the fundamental logic for modern computing.

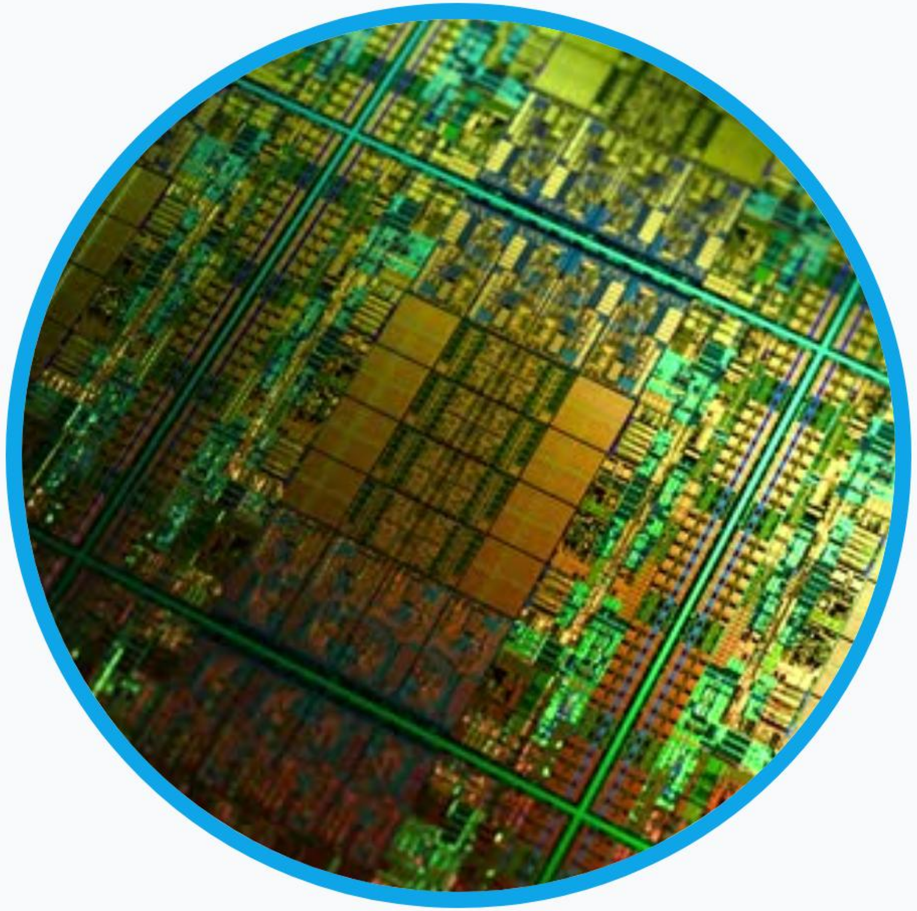
The Power: Proves that any problem solvable by an algorithm can be solved by a Turing Machine, regardless of hardware complexity.

Example: Imagine a roll of paper; a "head" reads a symbol, looks at a rulebook, and moves the tape. This is the ancestor of your CPU.



A fanciful mechanical Turing machine's TAPE and HEAD. The TABLE instructions might be on another "read only" tape, or perhaps on punch-cards. Usually a "finite state machine" is the model for the TABLE.

2. CPU (Central Processing Unit)



The "Brain" of the Computer

The primary component that fetches, decodes, and executes instructions from hardware and software.

- **Arithmetic & Logic:** Performs core calculations.
- **Control:** Orchestrates input/output operations.

When you click "Save," the CPU receives the command, calculates the data path, and instructs the drive to write.

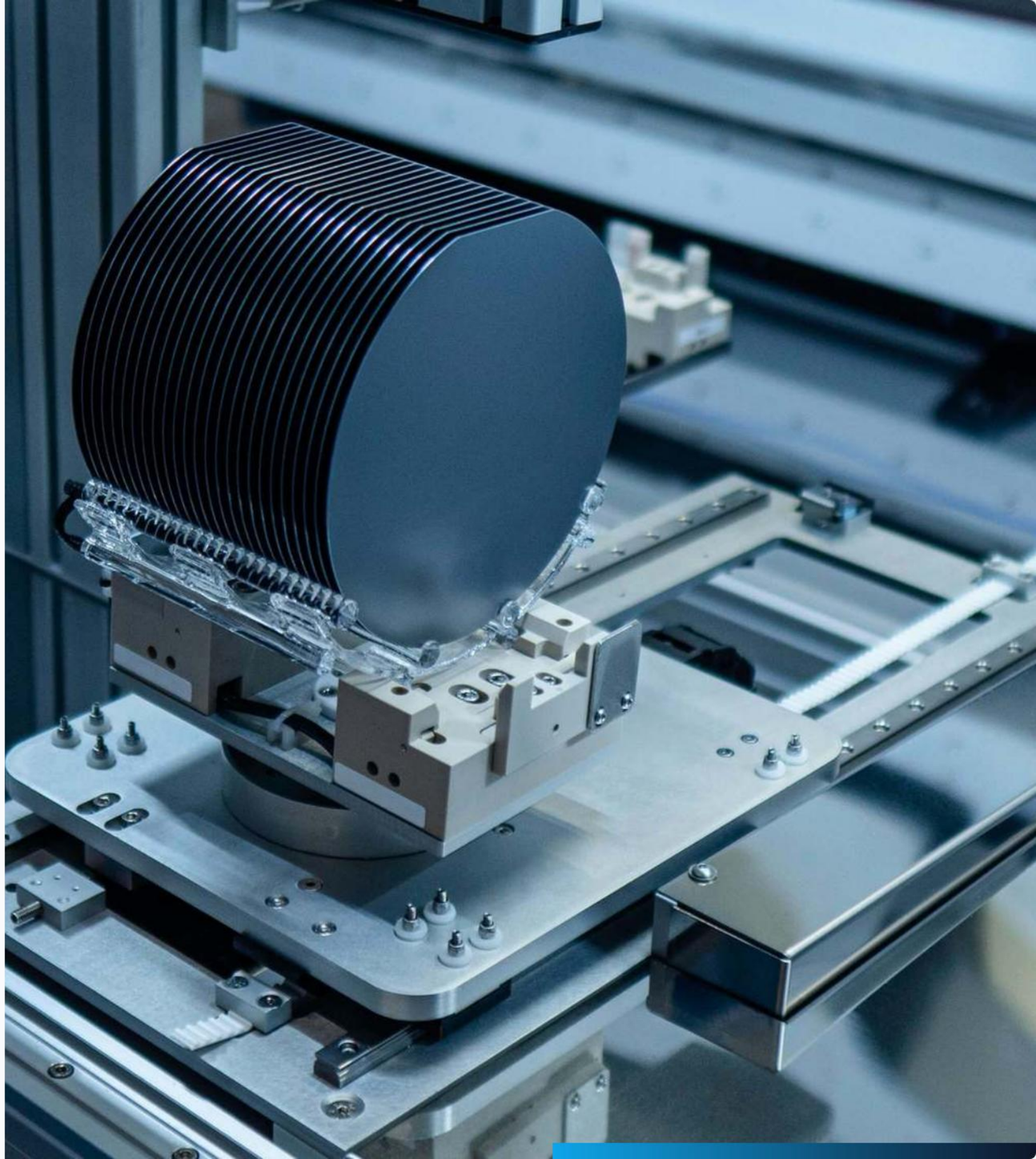
3. Silicon

The Foundation of Electronics

A semiconductor element that allows for precise control of electrical conductivity.

By adding impurities (doping), silicon acts as either a conductor or insulator, forming microscopic switches that process data.

Example: Think of it as the "wall" that holds millions of tiny electric switches together in a single chip.



4. Transistors

The Digital Switch

The fundamental building block of all digital electronics. A tiny semiconductor device acting as an electronic switch or amplifier.

By switching between "on" (1) and "off" (0), billions of transistors perform logic operations and store data.

The Faucet Analogy: Pressure on the handle (Gate) allows water (current) to flow. No pressure stops it. This on/off state represents one bit.

Electronic Transistors

5. Bit (Binary Digit)

0 / 1

The Indivisible Unit

The most fundamental unit of data in computing. It represents a logical state with one of two possible values.

Because computers run on electricity, a bit is either "on" (1) or "off" (0). It acts like a digital light switch.

Example: A Yes/No answer. "Is the computer plugged in?" 1 for Yes, 0 for No.

6. Byte

Definition

A unit consisting of 8 bits grouped together. The standard building block for measuring memory.

Capacity

Can represent 256 different combinations (2^8), enough to encode letters or symbols.

Example

Typing "A" stores as 01000001 in binary—exactly one byte of data.

7. Machine Code

The CPU's Native Tongue

The lowest-level software language, consisting entirely of 0s and 1s.
It is the only language a CPU understands directly without translation.

While we write `print("Hello")`, the computer sees `01001000 01100101...`

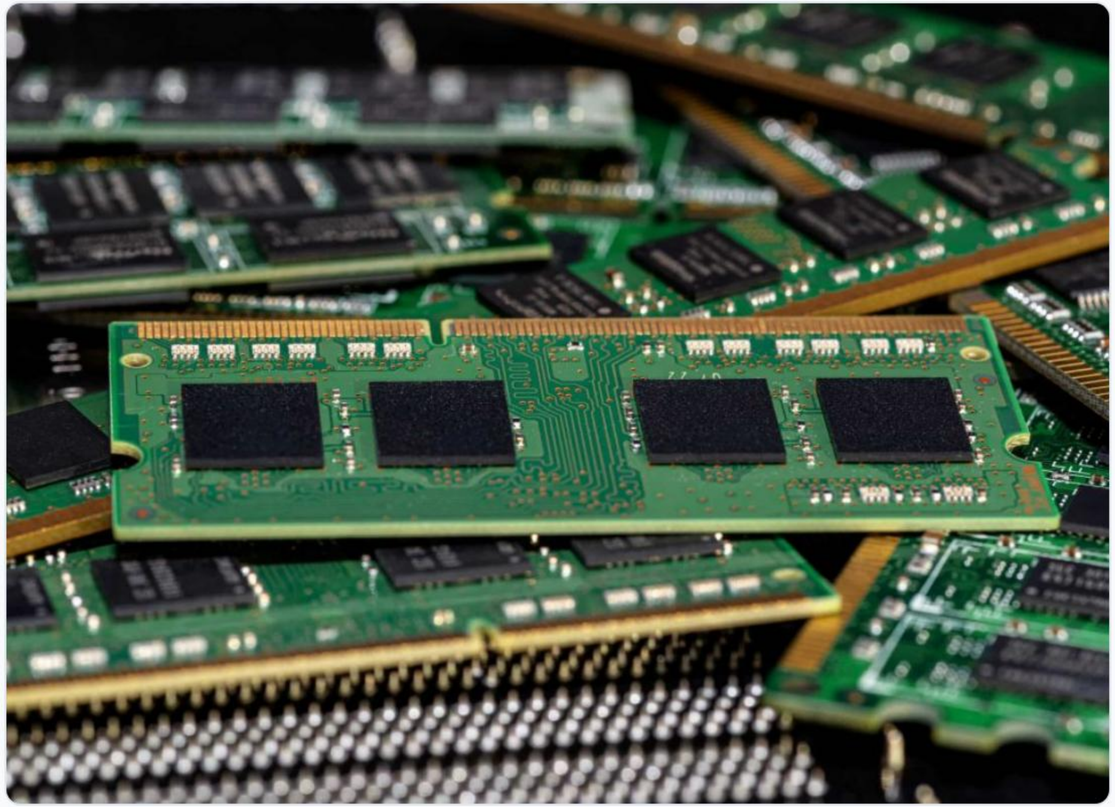
8. RAM (Random Access Memory)

High-Speed Workspace

Computer's high-speed, short-term memory. It stores active programs for near-instant CPU access.

Volatile Memory: RAM loses all data when power is turned off. It is a workspace, not a storage unit.

The Desk Analogy: The hard drive is like a slow filing cabinet; RAM is the surface of your desk where you keep current work.

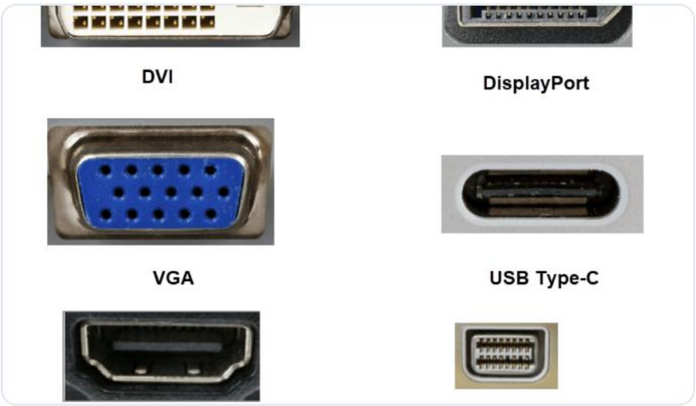


9. I/O (Input/Output)

The interface that allows humans to interact with digital logic. Input is data sent *to* the computer; Output is data sent *from* it.

Input (Keyboard)

Input (Mouse)



Output (Monitor)

10. Printers

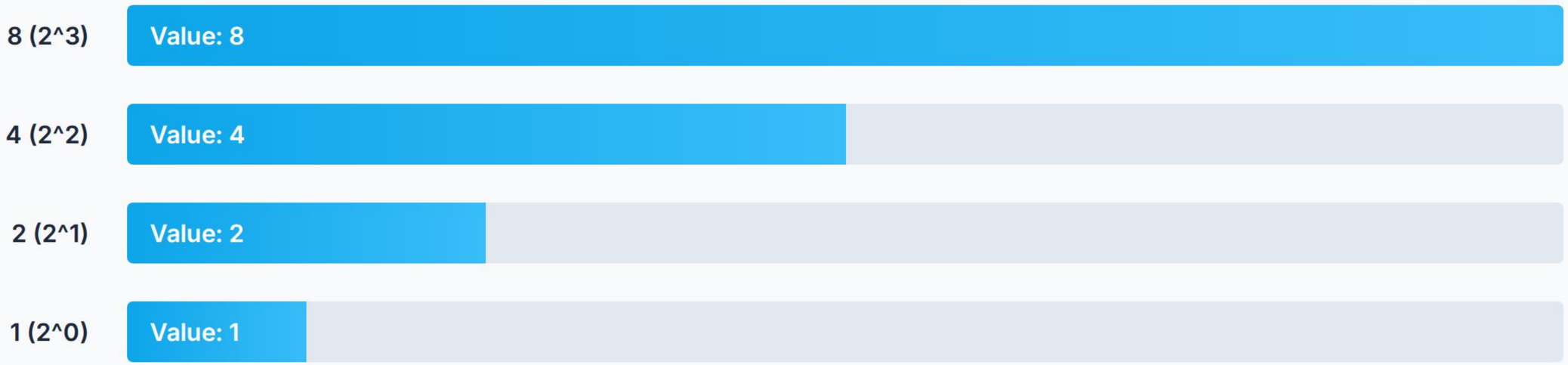
Digital to Physical

A peripheral that converts digital data into hard copies. Technologies include spraying ink (Inkjet) or using static electricity (Laser).

They are complex systems requiring perfect synchronization of mechanical parts, chemicals (toner), and software drivers.

Fun Fact: They are notorious for failing because of this high complexity across three different domains.

11. Binary (Base-2)



Binary uses place values that double with each position. The number 5 is **101** (one 4 + zero 2s + one 1).

12. Base-10 (Decimal)

- **Standard Human System:** Uses ten distinct digits (0-9).
- **Positional Value:** Each digit's value is based on powers of 10 (e.g., 256 = 200 + 50 + 6).
- **Biological Origin:** Likely adopted because humans have ten fingers for counting.

While computers "think" in binary (1010), humans conceptualize in Base-10 (10).

13. Hexadecimal (Base-16)

14. Nibble

13. Hexadecimal (Base-16)

14. Nibble

15. ASCII

16. UTF-8

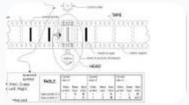
17. Big Endian

18. Little Endian

19. Kernel

20. Drivers

Image Sources



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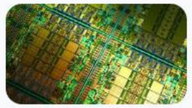

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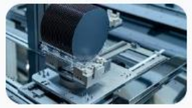

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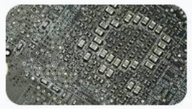

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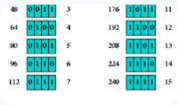

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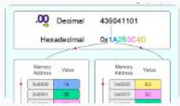

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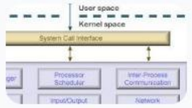

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